BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, FAISALABAD

INTERMEDIATE (12TH CLASS) EXAMINATION, 2021

BIOLOGY

Past Paper 2021

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Biology (Objective) Group - I

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8467

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. DNA polymerase enzyme which plays a supporting role in DNA replication is:

(A) Polymerase II
(B) Polymerase I
(C) Polymerase III
(D) Polymerase IV

Q2. The S-phase of cell cycle takes:

(A) 9 hours
(B) 4.5 hours
(C) 1.30 hours
(D) 10 hours

Q3. Pairing of homologous chromosomes called synapsis starts during:

(A) Leptotene
(B) Zygotene
(C) Pachytene
(D) Diakinesis

Q4. A person having neither antigen A nor B would have blood group:

(A) O
(B) A
(C) B
(D) AB

Q5. Organisms that have a foreign gene inserted into them are called:

(A) Transduct
(B) Transform
(C) Transgenic organism
(D) Bioreactors

Q6. Archaebacteria tolerate temperature up to:

(A) 10°C
(B) 40°C
(C) 120°C
(D) 140°C

Q7. Lithosphere includes:

(A) Air
(B) Water
(C) Gases
(D) Earth, soil

Q8. In grassland ecosystem, tropical climates have woody trees called:

(A) Savanna
(B) Pampas
(C) Prairies
(D) Alpine

Q9. The cause of acid rain is:

(A) Oxides of hydrogen
(B) NO_2 and SO_2
(C) Oxides of potassium
(D) Oxides of magnesium

Q10. Removal of salts with water from sweat glands and of sebum seems to be:

(A) Excretory
(B) Protective
(C) Thermo-regulation
(D) Both B & C

Q11. Kidneys receive what amount of blood supplied with each cardiac beat:

(A) 10 %
(B) 20 %
(C) 1 %
(D) 25 %

Q12. Long tubular structures join end to end to form long water conducting pipes in xylem are known as:

(A) Fibers
(B) Vessels
(C) Sclereids
(D) Trachea

Q13. Tropomyosin is a complex of how many polypeptide chains?

(A) Single
(B) Double
(C) Triple
(D) None

Q14. The receptors which have undifferentiated endings and produce sensation of pain are called:

(A) Chemo-receptors
(B) Nociceptors
(C) Mechano-receptors
(D) Thermo-receptors

Q15. Which is a haploid cell?

(A) Spermatogonia
(B) Primary spermatocyte
(C) Secondary spermatocyte
(D) Germinal epithelium

Q16. The final size of a given type of a cell is attained during:

(A) Maturation
(B) Differentiation
(C) Growth
(D) Elongation

Q17. The peripheral part of the blastoderm where the cells lie unseparated from the yolk is called:

(A) Hypoblast
(B) Epiblast
(C) Area pellucida
(D) Area opaca

Biology (Subjective) Group - I

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8467

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Define anhydrobiosis with an example.

(ii) What is glomerular filtrate?

(iii) What is pyrexia?

(iv) What is a ligament?

(v) Differentiate between hyaline cartilage and elastic cartilage.

(vi) How many ribs do not attach with the sternum?

(vii) What is after birth?

(viii) Define climacteric.

(ix) What is the productivity of grassland ecosystem?

(x) What are zooplankton? Give example.

(xi) Define eutrophication.

(xii) Give importance of forests.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Compare nerve impulse with saltatory impulse.

(ii) What is cerebrospinal fluid? Give its function.

(iii) What is acetylcholine? Give its role.

(iv) Differentiate between alleles and multiple alleles.

(v) What is universal blood donner?

(vi) What are opsins?

(vii) Give difference between ex-vivo and in-vivo gene therapy.

(viii) How hypercholesterolemia can be cured by gene therapy?

(ix) How cancer patients are being treated by gene therapy?

(x) Define biosphere.

(xi) Differentiate between habitat and ecological niche.

(xii) Define food chain. Give example.

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Compare morula and blastula.

(ii) How does coelom develop in chick embryo?

(iii) Compare heterochromatin and euchromatin.

(iv) Define transformation.

(v) Differentiate between template and coding strand of DNA.

(vi) Calculate the length of human cell cycle.

(vii) Compare kinetochore microtubules and polar microtubules.

(viii) How does molecular biology provide an evidence for evolution? Give at least one example.

(ix) Can migration affect the genotype frequency? If yes, how?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) How osmoregulation occurs in fresh water and terrestrial environment?

Q5(b). (Marks 04) Describe symbiosis and mutualism.

Q6(a). (Marks 04) Write the process of ecdysis in arthropods.

Q6(b). (Marks 04) Explain process of translation.

Q7(a). (Marks 04) What are receptors? Write names and functions of any four receptors

Q7(b). (Marks 04) What is greenhouse effect?

Q8(a). (Marks 04) Give an account of sexually transmitted diseases in man.

Q8(b). (Marks 04) Write note on mother-foetal Rh incompatibility.

Q9(a). (Marks 04) Discuss the Notochord and Mesoderm formation in chick embryo.

Q9(b). (Marks 04) Describe the evidences of evolution from comparative anatomy.

GROUP – II

Biology (Objective) Group - II

Time Allowed: 20 Minutes | **Maximum Marks:** 17 | **Paper Code:** 8466

Note: You have four choices for each objective type question as A, B, C and D. The choice which you think is correct, fill the relevant circle in front of that question number on computerized answer sheet. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero marks in that question. Attempt as many questions as given in objective type question paper and leave other circles blank.

Q1. Environmental factors causing abnormal development are called:

(A) Toxins
(B) Carcinogens
(C) Teratogens
(D) Mutagens

Q2. A group of cells that is capable of division is known as:

(A) Meristem
(B) Primordium
(C) Zone of cell division
(D) Zone of cell elongation

Q3. Parthenocarpy is artificially induced for commercial purpose by adding:

(A) Ethene
(B) Abscisic acid
(C) Cytokinins
(D) Auins

Q4. The earliest form of muscles to evolve was:

(A) Smooth muscles
(B) Cardiac muscles
(C) Skeletal muscles
(D) Voluntary muscles

Q5. Secondary walls of sclerenchyma cells are impregnated with:

(A) Cutin
(B) Suberin
(C) Pectin
(D) Lignin

Q6. A group of diseases in which bone resorption out paces bone deposit is known as:

(A) Osteoporosis
(B) Osteoarthritis
(C) Osteomalacia
(D) Arthritis

Q7. Nitrogen of amino acids is converted into urea by:

(A) Kidney
(B) Liver
(C) Spleen
(D) Pancreas

Q8. Humming bird is included in:

(A) Ectotherms
(B) Endotherms
(C) Heterotherms
(D) Poikilotherms

Q9. Which is green house gas:

(A) Oxygen
(B) Nitrogen
(C) Hydrogen
(D) Carbon dioxide

Q10. Northern coniferous forests are called:

(A) Taiga
(B) Savanna
(C) Prairies
(D) Tundra

Q11. The process in which micro-organism use proteins and release ammonia or ammonium ion is called:

(A) Nitrification
(B) Denitrification
(C) Ammonification
(D) Assimilation

Q12. A group of bacteria that tolerate temperature up to 120°C are called:

(A) Cyanobacteria
(B) Eubacteria
(C) Archaebacteria
(D) Mycoplasma

Q13. A gene is synthesized in laboratory from mRNA using:

(A) Reverse transcriptase
(B) DNA polymerase
(C) Transcriptase
(D) RNA polymerase

Q14. In a dihybrid cross the probability of plant with wrinkled and yellow seeds in F_2 is:

(A) 1/16
(B) 3/16
(C) 9/16
(D) 16/16

Q15. The spread of tumor cell and establishment of secondary areas of growth is called:

(A) Metamorphosis
(B) Cytostasis
(C) Epistasis
(D) Metastasis

Q16. The microtubules of spindle are composed of protein:

(A) Actin
(B) Myosin
(C) Globulin
(D) Tubulin

Q17. Synthesis of mRNA copy from DNA template is called:

(A) Transcription
(B) Translation
(C) Transduction
(D) Transformation

Biology (Subjective) Group - II

Time Allowed: 02:40 Hours | **Maximum Marks:** 68 | **Paper Code:** 8466

Note: SECTION - I

Q2. (Marks 16) Write short answers to any EIGHT parts.

(i) Define excretophore and anhydrobiosis.

(ii) Define panting with an example.

(iii) Compare osmoregulation in marine fishes and fresh water fishes.

(iv) Define moulting. Give the hormone involved.

(v) Give the structural composition of synovial joint.

(vi) What is osteomalacia and cleft palate?

(vii) Give the mechanism of invitro fertilization.

(viii) Define parthenocarpy and seed dormancy.

(ix) Give the layering feature of grassland.

(x) Write scientific names of any two animals of temperate deciduous forest.

(xi) Define greenhouse effect. Give its causes.

(xii) Define deforestation and afforestation.

Q3. (Marks 16) Write short answers to any EIGHT parts.

(i) Write any four junctions of cytokinins.

(ii) Differentiate between photoreceptors and thermoreceptors.

(iii) Write a note on Parkinson's disease.

(iv) Differentiate between genotype and phenotype.

(v) What do you understand by independent assortment of alleles?

(vi) Write a note on codominance.

(vii) What are palindromic sequences?

(viii) What is a nectar in biotechnology? Give its role.

(ix) What is gene pharming?

(x) Differentiate between ecosystem and biosphere.

(xi) Define food chain and food web.

(xii) What is denitrification?

Q4. (Marks 12) Write short answers to any SIX parts.

(i) Differentiate between primary and secondary growth.

(ii) Define teratology.

(iii) Write the function of DNA polymerase III.

(iv) What is meant by karyotype?

(v) Give the structure of typical nucleotide.

(vi) Differentiate between benign tumor and malignant tumor.

(vii) What is tetrad?

(viii) State theory of special creation.

(ix) What are vestigial organs?

> Note: SECTION - II Attempt any THREE questions. Each question carries 08 marks.

Q5(a). (Marks 04) Write a note on adaptations in plants to low and high temperature.

Q5(b). (Marks 04) What is predation? Write significance of predation.

Q6(a). (Marks 04) What are turgor movements? Describe their types.

Q6(b). (Marks 04) Describe the replication process of DNA.

Q7(a). (Marks 04) What are neurons? Explain their different types with the help of diagrams.

Q7(b). (Marks 04) Define acid rains. Describe their adverse effects.

Q8(a). (Marks 04) Write a note on test tube babies and identical twins.

Q8(b). (Marks 04) Define and explain multiple alleles with the help of example.

Q9(a). (Marks 04) Discuss any four factors affecting gene frequency.

Q9(b). (Marks 04) Write a note on regeneration.